

① Ideální plyn, He

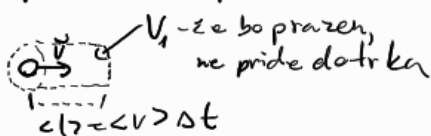
$$\langle W_k \rangle = ?$$

$$\langle l \rangle = ?$$

povp. w_k :

$$p = \frac{2}{3} \langle W_k \rangle \cdot n$$

povp. prostá pot



$$V_1 = \pi (2r)^2 \cdot \langle l \rangle$$

$$n = \frac{N}{V} = \frac{1}{V_1} \rightarrow V_1 = \frac{1}{n} = \frac{M}{m N_A} =$$

$$= \frac{M m R T k_B}{k N_A M p} \sim \frac{T k_B}{p} = 4 \pi r^2 \langle l \rangle$$

$$\langle l \rangle = \frac{T k_B}{4 \pi r^2 p} \quad \text{PŘIBLIŽNO}$$

$$\langle l \rangle = \frac{k_B T}{\sqrt{2} 4 \pi r^2 p} \quad \text{TOČNO}$$

$$pV = \frac{m}{M} RT$$

$$\frac{m}{M} \cdot \frac{RT}{p} = \frac{2}{3} \langle W_k \rangle \cdot \frac{1}{M} \frac{N_A}{V}$$

$$\langle W_k \rangle = \frac{3}{2} \frac{R}{N_A} \cdot T = \frac{3}{2} k_B T$$

$$\langle v \rangle = \sqrt{\frac{8 RT}{\pi M}}$$

$$\sqrt{\langle v^2 \rangle} = \sqrt{\frac{3 RT}{M}}$$

$$\langle v \rangle = \sqrt{\frac{8 RT}{\pi M}} = 1260 \text{ m/s}$$

$$\langle l \rangle = \frac{k_B T}{\sqrt{2} 4 \pi (2r)^2 p} = 0,2 \mu\text{m}$$

$$r \sim 10^{-10} \text{ m}$$

$$T = 300 \text{ K}$$

$$M = 4 \text{ kg/kmol}$$

$$p = 1 \text{ bar}$$

1.1.2 He

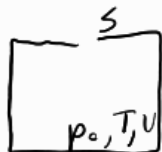
$$V = 1 \text{ l}$$

$$T = 300 \text{ K}$$

$$S = 10^{-3} \text{ m}^2$$

$$p_k = \frac{p_0}{2}$$

$$t = ?$$



$$pV = \frac{N}{N_A} RT$$

$$\rightarrow N = \frac{N_A p_0 V}{RT}$$

$$\Delta N = \frac{1}{4} n S v \Delta t = \frac{1}{4} \frac{m N_A}{M V} S \langle v \rangle \Delta t$$

$$\frac{\Delta N}{\Delta t} = \frac{1}{4} \frac{m N_A}{M V} \phi_v \quad ; \quad \frac{m}{M V} = \frac{p}{RT}$$

$$\frac{\Delta N}{\Delta t} = \frac{1}{4} \frac{p N_A}{RT} \phi_v$$

$$-dN_{\text{vpodsti}} = \left(\frac{\Delta N_{\text{vch}}}{\Delta t} - \frac{\Delta N_{\text{vot}}}{\Delta t} \right) dt = \frac{1}{4} \frac{N_A}{RT} (p_1 - p_0) S \langle v \rangle dt$$

$$-\frac{N_A V}{RT} dp_1 = \frac{1}{4} \frac{N_A}{RT} (p_1 - p_0) \langle v \rangle dt$$

$$-\int_{p_1}^p \frac{1}{p_1 - p_0} dp_1 = \int_0^t \frac{1}{4} \frac{\langle v \rangle}{V} dt$$

$$-\ln \frac{p - p_0}{p_1 - p_0} = \frac{1}{4} \frac{\langle v \rangle}{V} t$$

$$t = -\frac{4V}{\langle v \rangle} \ln \frac{p - p_0}{p_1 - p_0} \xrightarrow{p \rightarrow \infty} \boxed{t = \frac{4V}{\langle v \rangle} \ln 2}$$

$$\frac{p - p_0}{p_1 - p_0} = e^{-\frac{\langle v \rangle t}{4V}}$$

$$p = p_0 + \underbrace{(p_1 - p_0)}_{\Delta p \text{ na začátku}} e^{-\frac{\langle v \rangle t}{4V}}$$